

Tecnológico de Monterrey

Nombre del Proyecto

Etapa 1: Conociendo el negocio

Equipo 1

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1. Descripción de los datos

Para esta primera etapa, empezaremos por cargar las librerías que necesitamos y establecemos algunas opciones de configuración:

```
library(tidyverse)
library(here)
library(readxl)

options(scipen = 999)
```

Ahora cargamos los datos que nos proporcionó el Socio Formador. Para los datos del 2025 tenemos:

```
calidad_aire_2025_tbl <- read_excel(here("data", "BD 2025.xlsx")) |>
  glimpse()
```

```
Rows: 4,344
Columns: 16
$ date  <dttm> NA, 2025-01-01 00:00:00, 2025-01-01 01:00:00, 2025-01-01 02:00:~
$ CO    <chr> "ppm", NA, "0.18", "0.18", "0.19", "0.18", "0.17", "0.17", "0.17~
$ NO    <chr> "ppb", NA, "2.6", "2.8", "3.1", "2.8", "2.6", "2.6", "3", "8.1",~
$ NO2   <chr> "ppb", NA, "7.5", "13.5", "15.1", "12.8", "4.8", "4.2", "9.80000~
$ NOX   <chr> "ppb", NA, "10.1", "16.2", "18.100000000000001", "15.6", "7.4", ~
$ O3    <chr> "ppb", "36", "37", "26", "22", "24", "33", "36", "21", "9", "9",~
$ PM10  <dbl> NA, 104, 58, 79, 66, 48, 46, 35, 41, 42, 43, 43, 80, 51, 47, 47,~
$ PM2.5 <dbl> NA, 31.00, 30.84, 49.92, 45.87, 35.65, 31.14, 27.43, 25.33, 28.4~
$ PRS   <chr> "mmHg", NA, "725.1", "725.1", "725.1", "725.1", "725.5", "725.9"~
$ RAINF <chr> "mmh", NA, "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0"~
$ RH    <chr> "%", NA, "66", "66", "74", "76", "86", "87", "85", "86", "84", "~
$ SO2   <chr> "ppb", NA, "4.0999999999999996", "4.3", "4.0999999999999996", "4~
$ SR    <chr> "kW/m2", NA, "0", "0", "0", "0", "0", "0", "0", "6.0000000000000~
$ TOUT  <chr> "°C", NA, "17.62", "16.579999999999998", "15.5", "15.03", "14.32~
$ WSR   <chr> "km/h", NA, "10.7", "5.3", "5.3", "5", "8.6999999999999993", "7.~
$ WDR   <chr> "Azimutal", NA, "136", "255", "250", "181", "91", "104", "263", ~
```

Para los datos del 2024 tenemos:

```
calidad_aire_2024_tbl <- read_excel(here("data", "BD 2024.xlsx")) |>
  glimpse()
```

```
Rows: 8,784
Columns: 16
$ `Fecha y hora`   <dttm> 2024-01-01 00:00:00, 2024-01-01 01:00:00, 2024-01-01~
$ `CO (ppm)`       <dbl> 2.08, 2.07, 2.36, 2.11, 1.94, 1.79, 1.59, 1.53, 1.65,~
$ `NO (ppb)`       <dbl> 10.5, 10.5, 16.9, 13.6, 8.5, 7.4, 7.4, 7.0, 8.4, 6.1,~
$ `NO2 (ppb)`      <dbl> 23.3, 23.4, 30.0, 26.6, 20.6, 16.3, 14.2, 12.3, 19.8,~
$ `NOX (ppb)`      <dbl> 33.8, 33.9, 46.9, 40.1, 29.2, 23.6, 21.5, 19.2, 28.2,~
$ `O3 (ppb)`       <dbl> 13, 13, 7, 7, 13, 15, 21, 22, 18, 37, 42, 45, 45, 48,~
```

```
$ `PM10 (ug/m3)`   <dbl> 81, 81, 84, 132, 93, 62, 42, 26, 29, 40, 28, 28, 30, ~
$ `PM2.5 (ug/m3)`  <dbl> 60.00, 60.52, NA, 64.69, 44.13, 28.90, 16.64, 12.57, ~
$ `PRS (mmHg)`     <dbl> 724.5, 724.5, 724.6, 724.7, 724.7, 724.7, 724.7, 725.~
$ `RAINF (mm/h)`   <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
$ `RH (%)`         <dbl> 58, 58, 61, 62, 53, 55, 42, 43, 48, 43, 35, 33, 36, 3~
$ `SO2 (ppb)`      <dbl> 2.9, 2.9, 3.2, 3.2, 3.2, 2.9, 3.2, 2.9, 3.7, 6.2, 6.4~
$ `SR (kW/m2)`     <dbl> 0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.00~
$ `TOUT (°C)`      <dbl> 14.26, 14.28, 13.50, 13.18, 14.62, 13.96, 16.39, 15.9~
$ `WSR (km/h)`     <dbl> 5.8, 5.9, 3.7, 7.1, 8.4, 7.6, 9.7, 9.6, 5.8, 5.2, 6.3~
$ `WDR (azimutal)` <dbl> 290, 289, 291, 291, 297, 292, 283, 291, 223, 164, 163~
```

Para los datos del 2023 tenemos:

```
calidad_aire_2023_tbl <- read_excel(here("data", "BD 2023.xlsx")) |>
  glimpse()
```

```
Rows: 8,758
Columns: 16
$ `Fecha y hora` <dttm> 2023-01-01 00:00:00, 2023-01-01 01:00:00, 2023-01-01 0~
$ CO             <dbl> 1.67, 1.70, 1.51, 1.30, 1.28, 2.12, 1.54, 1.04, 0.98, 1~
$ NO             <dbl> 4.1, 5.0, 4.3, 3.8, 4.6, 19.8, 7.0, 3.8, 3.7, 6.5, 6.2,~
$ NO2            <dbl> 30.3, 28.9, 23.2, 20.9, 20.7, 30.6, 23.7, 13.6, 12.3, 1~
$ NOX            <dbl> 34.5, 34.1, 27.7, 24.9, 25.5, 50.6, 30.8, 17.6, 16.3, 2~
$ O3             <dbl> 9, 9, 11, 11, 9, 4, 7, 16, 19, 19, 25, 28, 34, 33, 47, ~
$ PM10           <dbl> 118, 97, 103, 83, 65, 67, 147, 84, 43, 36, 41, 44, 33, ~
$ PM2.5          <dbl> 94.12, 96.79, 81.23, 49.83, 40.19, NA, 108.13, 35.00, 1~
$ PRS            <dbl> 718.4, 710.7, 710.4, 710.1, 710.2, 710.2, 710.2, 710.2,~
$ RAINF          <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
$ RH             <dbl> 45, 47, 46, 48, 55, 61, 55, 43, 39, 33, 27, 21, 18, 17,~
$ SO2            <dbl> 3.3, 3.4, 3.4, 3.2, 3.0, 3.1, 3.0, 3.3, 3.2, 3.1, 3.1, ~
$ SR             <dbl> 0.000, 0.001, 0.001, 0.001, 0.001, 0.001, 0.001, 0.001,~
$ TOUT           <dbl> 17.65, 17.12, 16.69, 15.99, 14.74, 14.53, 14.55, 16.08,~
$ WSR            <dbl> 4.9, 3.4, 3.4, 2.6, 3.9, 3.4, 3.4, 7.1, 6.3, 3.5, 4.1, ~
$ WDR            <dbl> 236, 336, 3, 315, 270, 242, 307, 232, 222, 18, 294, 94,~
```

Para los datos del 2022 tenemos:

```
calidad_aire_2022_tbl <- read_excel(here("data", "BD 2022.xlsx")) |>
  glimpse()
```

```
Rows: 8,760
Columns: 16
$ `Fecha y hora` <dttm> 2022-01-01 00:00:00, 2022-01-01 01:00:00, 2022-01-01 0~
$ CO             <dbl> 2.60, 2.23, 1.99, 2.03, 1.86, 1.76, 1.55, 1.41, 1.31, 1~
$ NO             <dbl> 3.5, 2.9, 2.9, NA, NA, NA, 2.8, 2.9, 3.0, 6.6, 17.1, 5.~
$ NO2            <dbl> 44.9, 32.9, 27.6, NA, NA, NA, 16.4, 13.5, 13.1, 16.5, 2~
$ NOX            <dbl> 48.5, 36.0, 30.7, NA, NA, NA, 19.4, 16.5, 16.3, 23.4, 4~
$ O3             <dbl> 15, 19, 21, 18, 20, 20, 22, 21, 17, 13, 12, 26, 30, 39,~
```

1 Descripción de los datos

```
$ PM10           <dbl> 134, 141, 117, 108, 106, 81, 71, 59, 57, 52, 54, 68, 62~
$ PM2.5          <dbl> 91.00, 112.61, 92.46, 69.20, 68.03, 49.64, 39.01, 26.40~
$ PRS            <dbl> 705.5, 705.2, 705.1, 704.8, 704.8, 704.7, 704.9, 705.3,~
$ RAINF          <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
$ RH             <dbl> 46, 48, 46, 46, 46, 42, 37, 31, 26, 27, 23, 18, 17, 16,~
$ SO2            <dbl> 5.3, 6.1, 5.8, 6.0, 5.6, 5.3, 4.9, 4.6, 4.1, 3.9, 4.1, ~
$ SR             <dbl> 0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000,~
$ TOUT           <dbl> 21.37, 20.83, 20.52, 20.12, 19.85, 19.80, 20.05, 21.15,~
$ WSR            <dbl> 3.4, 3.3, 5.9, 5.1, 4.7, 4.2, 4.3, 5.1, 5.9, 4.2, 4.5, ~
$ WDR            <dbl> 267, 259, 233, 220, 244, 132, 235, 222, 209, 97, 148, 1~
```

Para los datos del 2021 tenemos:

```
calidad_aire_2021_tbl <- read_excel(here("data", "BD 2021.xlsx")) |>
  glimpse()
```

```
Rows: 8,760
Columns: 16
$ `Fecha y hora` <dttm> 2021-01-01 00:00:00, 2021-01-01 01:00:00, 2021-01-01 0~
$ CO             <dbl> NA, 4.73, 4.72, 4.82, 5.02, 4.79, 4.77, 4.73, 4.76, 4.6~
$ NO             <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
$ NO2            <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
$ NOX            <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
$ O3             <dbl> NA, 27, 23, 25, 25, 19, 20, 17, 13, 15, 20, 20, 20, 24,~
$ PM10           <dbl> NA, 68, 55, 57, 65, 71, 60, 57, 47, 45, 30, 39, 41, 37,~
$ PM2.5          <dbl> NA, 46.00, 47.64, 52.07, 61.62, 49.13, 48.03, 37.10, 29~
$ PRS            <dbl> NA, 709.7, 710.1, 710.4, 710.6, 710.6, 710.8, 711.3, 71~
$ RAINF          <dbl> NA, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ RH             <dbl> NA, 67, 70, 73, 74, 76, 78, 77, 78, 76, 69, 63, 61, 57,~
$ SO2            <dbl> NA, 3.3, 3.3, 3.6, 3.5, 3.6, 3.4, 3.5, 3.3, 3.3, 3.9, 4~
$ SR             <dbl> NA, 0.167, 0.165, 0.164, 0.164, 0.163, 0.163, 0.163, 0.~
$ TOUT           <dbl> NA, 8.26, 7.15, 6.20, 5.64, 5.08, 4.73, 4.68, 4.39, 4.6~
$ WSR            <dbl> NA, 7.9, 4.8, 5.2, 3.2, 3.0, 2.4, 3.0, 4.7, 4.5, 4.7, 6~
$ WDR            <dbl> NA, 78, 47, 349, 358, 19, 8, 16, 14, 10, 36, 43, 42, 61~
```

Finalmente, para los datos del 2024 tenemos:

```
calidad_aire_2020_tbl <- read_excel(here("data", "BD 2020.xlsx")) |>
  glimpse()
```

```
Rows: 8,779
Columns: 16
$ `Fecha y hora` <dttm> 2020-01-01 00:00:00, 2020-01-01 01:00:00, 2020-01-01 0~
$ CO             <dbl> NA, 2.11, 2.06, 1.96, 1.98, 2.06, 1.94, 2.06, 1.94, 2.0~
$ NO             <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
$ NO2            <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
$ NOX            <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
$ O3             <dbl> NA, 19, 19, 19, 16, 13, 15, 10, 12, 7, 8, 9, 10, 8, 16,~
```

```
$ PM10           <dbl> 66, 57, 68, 68, 48, 42, 38, 33, 36, 21, 35, 37, 31, 28,~
$ PM2.5          <dbl> 54.23, NA, 53.84, 36.47, 33.59, 33.42, 20.20, 23.84, 15~
$ PRS            <dbl> NA, 735.7, 734.8, 734.2, 733.9, 733.8, 733.8, 733.6, 73~
$ RAINF          <dbl> NA, 0.00, 0.00, 0.00, 0.00, 0.02, 0.10, 0.06, 0.02, 0.0~
$ RH             <dbl> NA, 96, 96, 96, 96, 96, 96, 96, NA, NA, NA, NA, 95, 95,~
$ SO2            <dbl> NA, 5.4, 5.5, 5.4, 5.5, 5.4, 5.4, NA, NA, NA, NA, NA, N~
$ SR             <dbl> 0.000, 0.010, 0.010, 0.010, 0.010, 0.010, 0.010, 0.010,~
$ TOUT           <dbl> NA, 11.20, 11.26, 11.35, 11.47, 11.69, 11.71, 11.84, 11~
$ WSR            <dbl> NA, 8.1, 5.5, 3.8, 3.3, 3.5, 2.9, 3.8, 3.0, 11.2, 13.0,~
$ WDR            <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
```